

CERTIFIED MATERIAL TEST REPORT

Type 3.1/3.1.B in accordance with EN10204:2004 & ISO 10474:2013

SOLD TO:

 EDGEN MURRAY
 18444 HIGHLAND RD
 BATON ROUGE LA 70809

SHIP TO:

 EDGEN MURRAY C/O MWPC
 925 KENNEDY AVENUE
 SCHERERVILLE IN 46375

CUSTOMER PO #

STL - 305994

SALES ORDER #

1359727

DELIVERY #

82521735

DATE OF CREATION

04/22/2024

PLANT

Warren - Warren , OH

Heat Analysis

Heat/Run	Value	Item Description	C	Mn	P	S	Cu	Ni	Cr	Mo	V	Si	Al	Nb	Ti	B
Heat	302586	4.500 .337 API5L X52M PSL2 HFW PEB BARE	0.22	0.73	0.008	0.002	0.080	0.030	0.070	0.010	0.002	0.030	0.030	0.001	0.001	0.0001
Heat	302599	4.500 .337 API5L X52M PSL2 HFW PEB BARE	0.20	0.72	0.008	0.002	0.100	0.050	0.080	0.020	0.003	0.010	0.020	0.001	0.001	0.0001

Tensile Test Results

Heat/Run	Value	Item Description	Test Date	Y/S	T/S	EI	HRB	Y/T Ratio
Heat	302586	4.500 .337 API5L X52M PSL2 HFW PEB BARE	04/12/2024	073414	080699	27	87	0.91
Heat	302599	4.500 .337 API5L X52M PSL2 HFW PEB BARE	04/12/2024	066396	075104	33	88	0.88

Product Analysis

Heat/Run	Description	C	Mn	P	S	Cu	Ni	Cr	Mo	V	Si	Al	Nb	Ti	B	CE
302586	4.500 .337 API5L X52M PSL2 HFW PEB BARE	0.22	0.72	0.010	0.005	0.080	0.033	0.070	0.014	0.000	0.028	0.034	0.000	0.001	0.0000	0.367
302586	4.500 .337 API5L X52M PSL2 HFW PEB BARE	0.22	0.74	0.006	0.003	0.076	0.032	0.072	0.014	0.001	0.030	0.038	0.000	0.002	0.0000	0.369
302599	4.500 .337 API5L X52M PSL2 HFW PEB BARE	0.21	0.72	0.009	0.003	0.094	0.052	0.083	0.016	0.001	0.017	0.023	0.000	0.001	0.0000	0.359
302599	4.500 .337 API5L X52M PSL2 HFW PEB BARE	0.20	0.73	0.009	0.004	0.097	0.054	0.085	0.017	0.002	0.017	0.024	0.000	0.002	0.0000	0.355

Manufacturing Locations

Heat/Run	Pipe Manufacturing	Coil Rolling	Steelmaking
302586	Warren -Warren,OH	NORTH STAR -DELTA,OH	NORTH STAR -DELTA,OH
302599	Warren -Warren,OH	NORTH STAR -DELTA,OH	NORTH STAR -DELTA,OH

COMMENTS:

This is to certify that the product described herein was manufactured or supplied by the Wheatland Tube Company, a division of Zekelman Industries, and have been sampled, tested and/or inspected in accordance with the latest revision, at the time of manufacture, of the specification listed below and fulfills the requirements in such respects. All the required mechanical, physical, hydrostatic, non-destructive, flattening, and bend tests have been successfully completed. The products described above meet the following specifications: API Specification 5L 46th Edition, with API monogram, ASTM A53/A53M 2020 and ASME SA53 2015, Type E, Grade B, the applicable hardness requirements of NACE MR0175.
 2.375x218 X52 PSL2 does not meet elongation for ASTM A53/ASME SA53

API SPECIFICATION 5L REQUIREMENTS:

Minimum hydrostatic test pressure (psi): 2970 for 5 seconds
 Electric-welded (HFW) product
 Weld seam annealing at 1600 deg. F. Minimum
 Ultrasonically tested (N-10 notch)
 Flattening test passed
 1 inch L90 strip specimen (1L90)
 For Carbon mass fraction greater than 0.12 %, maximum CE_{max} = 0.43;
 For Carbon mass fraction equal to or less than 0.12 %, maximum CE_{Pcm} = 0.25
 Delivery Condition: M
 Exempt from CVN impact testing per API Spec 5L, Table 22
 Formerly ENERGEX TUBE


Tiffany Dunworth
 Technical Services Manager

M&M USA

"M&M USA" designation denotes materials that are 100% melted and manufactured in the USA.

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PLANT

Warren - Warren , OH

Heat Analysis

Heat/Run	Value	Item Description	C	Mn	P	S	Cu	Ni	Cr	Mo	V	Si	Al	Nb	Ti	B
Heat	302590	4.500 .337 API5L X52M PSL2 HFW PEB BARE	0.20	0.71	0.009	0.004	0.080	0.030	0.080	0.020	0.002	0.020	0.020	0.001	0.001	0.0001
Heat	302597	4.500 .337 API5L X52M PSL2 HFW PEB BARE	0.21	0.72	0.008	0.003	0.080	0.050	0.070	0.020	0.001	0.010	0.030	0.001	0.001	0.0001
Heat	302599	4.500 .337 API5L X52M PSL2 HFW PEB BARE	0.20	0.72	0.008	0.002	0.100	0.050	0.080	0.020	0.003	0.010	0.020	0.001	0.001	0.0001
Heat	EA5324	4.500 .337 API5L X52M PSL2 HFW PEB BARE	0.19	0.68	0.013	0.002	0.120	0.040	0.040	0.020	0.002	0.030	0.027	0.000	0.001	0.0002

Tensile Test Results

Heat/Run	Value	Item Description	Test Date	Y/S	T/S	EI	HRB	Y/T Ratio
Heat	302590	4.500 .337 API5L X52M PSL2 HFW PEB BARE	04/11/2024	069829	079517	31	86	0.88
Heat	302597	4.500 .337 API5L X52M PSL2 HFW PEB BARE	04/11/2024	068118	076011	30	88	0.90
Heat	302599	4.500 .337 API5L X52M PSL2 HFW PEB BARE	04/12/2024	066396	075104	33	88	0.88
Heat	EA5324	4.500 .337 API5L X52M PSL2 HFW PEB BARE	04/11/2024	073299	082601	29	88	0.89

Product Analysis

Heat/Run	Description	C	Mn	P	S	Cu	Ni	Cr	Mo	V	Si	Al	Nb	Ti	B	CE
302590	4.500 .337 API5L X52M PSL2 HFW PEB BARE	0.20	0.72	0.009	0.005	0.076	0.033	0.081	0.017	0.001	0.022	0.028	0.000	0.001	0.0000	0.351
302590	4.500 .337 API5L X52M PSL2 HFW PEB BARE	0.21	0.71	0.010	0.005	0.075	0.032	0.080	0.017	0.001	0.022	0.028	0.000	0.001	0.0000	0.354
302597	4.500 .337 API5L X52M PSL2 HFW PEB BARE	0.22	0.71	0.010	0.005	0.080	0.050	0.067	0.016	0.000	0.019	0.031	0.000	0.002	0.0000	0.360
302597	4.500 .337 API5L X52M PSL2 HFW PEB BARE	0.21	0.73	0.008	0.003	0.079	0.048	0.068	0.016	0.000	0.016	0.031	0.000	0.001	0.0000	0.353
302599	4.500 .337 API5L X52M PSL2 HFW PEB BARE	0.21	0.72	0.009	0.003	0.094	0.052	0.083	0.016	0.001	0.017	0.023	0.000	0.001	0.0000	0.359
302599	4.500 .337 API5L X52M PSL2 HFW PEB BARE	0.20	0.73	0.009	0.004	0.097	0.054	0.085	0.017	0.002	0.017	0.024	0.000	0.002	0.0000	0.355
EA5324	4.500 .337 API5L X52M PSL2 HFW PEB BARE	0.20	0.76	0.009	0.002	0.109	0.039	0.040	0.012	0.004	0.030	0.031	0.001	0.000	0.0000	0.343
EA5324	4.500 .337 API5L X52M PSL2 HFW PEB BARE	0.20	0.77	0.012	0.002	0.111	0.041	0.040	0.013	0.003	0.032	0.030	0.001	0.001	0.0000	0.350

Manufacturing Locations

Heat/Run	Pipe Manufacturing	Coil Rolling	Steelmaking
302590	Warren -Warren,OH	NORTH STAR -DELTA,OH	NORTH STAR -DELTA,OH
302597	Warren -Warren,OH	NORTH STAR -DELTA,OH	NORTH STAR -DELTA,OH
302599	Warren -Warren,OH	NORTH STAR -DELTA,OH	NORTH STAR -DELTA,OH
EA5324	Warren -Warren,OH	NUCOR STEEL GALLATIN USA -GHENT,KY	NUCOR STEEL GALLATIN USA -GHENT,KY

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Tiffany Dunworth
Technical Services Manager

M&M USA

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